Lexicographic Constraint Programming for Recreational Softball Defensive Lineups: A Team Red Case Study

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Abstract

We study the construction of seven-inning defensive lineups for recreational slowpitch softball when the available roster exceeds the number of fielding positions. The problem requires a legal assignment in every inning while balancing playing time, respecting player-reported positional preferences, and limiting disruptive field-position and bench transitions. We formulate the problem as a constraint-programming model with a strict lexicographic objective: playing-time inequity is minimized first, followed by non-explicit assignments and several measures of lineup continuity. Eligibility distinguishes explicitly selected positions from directionally inferred alternatives, while pitcher and first base remain opt-in assignments. The model is implemented in OR-Tools CP-SAT through bounded, status-aware phases suitable for interactive use. For the fourteen-player Team Red snapshot studied here, the minimum playing-time spread is three innings and the minimum scaled absolute deviation is 56. A recorded legal incumbent fills all 70 defensive slots without a non-explicit assignment, uses no more than two field positions per player, and has continuity vector $(3, 0, 15, 20, 20)$. Its application status is *FEASIBLE*: the fairness and fallback values are certified, but complete lexicographic optimality of the continuity vector is not claimed.

Keywords: sports scheduling; constraint programming; CP-SAT; lexicographic optimization; playing-time fairness; human-centered operations

1 Introduction

Defensive scheduling in recreational slowpitch softball is a compact but nontrivial assignment problem. A conventional defense contains ten positions: pitcher (P), catcher (C), first base (1B), second base (2B), third base (3B), shortstop (SS), left field (LF), left center field (LC), right center field (RC), and right field (RF). A roster commonly contains more available players than defensive positions. The resulting schedule must repeatedly decide who fields, who occupies each position, and who is on the Bench.

The Open league considered here does not divide players by gender and does not impose gender minima. That simplification removes one class of hard constraints, but it does not make the practical objective obvious. Equal playing time is important; explicit positional preferences should be honored; players should not be moved unnecessarily; and a player who sits should, when possible, sit in a coherent block rather than alternate between the field and the Bench. These goals conflict. A smoother schedule is not necessarily a more preference-respecting schedule, and a perfectly continuous schedule is not necessarily fair.

The implementation resolves those conflicts by priority rather than by an opaque weighted compromise. The order is substantive: legality precedes fairness, fairness precedes aggregate non-explicit assignment count, and both precede continuity. Lower-priority improvements are prohibited from purchasing a worse higher-priority result. This formulation is particularly appropriate for an interactive tool, where a user should not have to interpret a hidden exchange rate between one inning of inequity and one position change.

This work makes three contributions:

1. it formulates recreational defensive scheduling as a finite-horizon state-assignment problem with an explicit bench state;
2. it separates preference declarations from directionally inferred eligibility and orders fairness, non-explicit assignment, and continuity objectives lexicographically; and
3. it develops a bounded CP-SAT procedure that preserves certified higher-priority values while returning a legal incumbent under an interactive time budget.

The presentation follows standard assignment-modeling practice [5, 6], but the repeated within-game assignment, directed eligibility expansion, and explicit bench state produce an application-specific multiobjective structure. Mathematical propositions establish structural and instance-specific bounds; solver statuses certify only those objective tiers proved optimal; regression tests check implementation conformance; and timed experiments characterize, rather than guarantee, latency and incumbent quality.

2 Operational Assumptions

The model optimizes one game at a time under the following assumptions.

- A regulation planning horizon contains seven innings.
- Availability is constant throughout the game. Late arrival and early departure are outside the current input model.
- Between eight and fifteen players may be selected as available. A stored roster may be larger, but a single solve is capped at fifteen.
- Every active position is filled in every modeled inning, and one player cannot occupy two simultaneous positions.
- Gender has no legal or objective effect in the Open profile. Team Red therefore stores an internal `Unspecified` sentinel and exposes no gender control.
- A checked position is an unranked explicit preference. Derived positions are legal fallbacks, not additional preferences.
- Every player may catch. Pitcher and first base require explicit opt-in because neither is inferred from another position.
- The Bench state means only that a player is not fielding in that inning. It does not remove the player from the batting order or model an offensive substitution.
- For a given confirmed input snapshot, the returned schedule is authoritative. The designated lineup caller may alter the inputs and solve again, but no approval gate or hand edit modifies the output.

Let N denote the number of available players. The active position set $\mathcal{S}(N)$ and its cardinality $K(N) = |\mathcal{S}(N)|$ are fixed before optimization according to Table 1.

An input with fewer than eight or more than fifteen available players is rejected before the constraint model is constructed. If ten or more players are available, the optimizer does not silently reduce the defense to nine to work around insufficient eligibility; it reports the eligibility failure.

Table 1: Active defensive positions under the Open rules profile.

Available players N	$K(N)$	$\mathcal{S}(N)$
$10 \le N \le 15$	10	P, C, 1B, 2B, 3B, SS, LF, LC, RC, RF
$N = 9$	9	all standard positions except RF
$N = 8$	8	all standard positions except C and RF

3 Preference and Eligibility Semantics

Let the complete position universe be

$$\mathcal{S} = \{\mathrm{P}, \mathrm{C}, 1\mathrm{B}, 2\mathrm{B}, 3\mathrm{B}, \mathrm{SS}, \mathrm{LF}, \mathrm{LC}, \mathrm{RC}, \mathrm{RF}\},$$

and let $\varnothing \neq Q_p \subseteq \mathcal{S}$ be player p's explicitly selected positions. Define the direct eligibility-expansion map H_f by

$$\begin{aligned} &H_f(\mathrm{SS}) = \{3\mathrm{B}, 2\mathrm{B}\}, && H_f(3\mathrm{B}) = \{2\mathrm{B}\}, \\ &H_f(\mathrm{LC}) = \{\mathrm{LF}, \mathrm{RC}, \mathrm{RF}\}, && H_f(\mathrm{LF}) = \{\mathrm{RC}, \mathrm{RF}\}, \\ &H_f(\mathrm{RC}) = \{\mathrm{RF}\}, && H_f(s) = \varnothing \quad \text{otherwise.} \end{aligned}$$

Thus selecting SS, for example, implies eligibility at 3B and 2B; the notation denotes eligibility expansion, not preference ranking or dominance. The map is applied directly rather than transitively: $h(Q_p) = \bigcup_{s \in Q_p} H_f(s)$. Every eligible non-explicit assignment—whether produced by H_f, universal catcher eligibility, or the eight-player rule below—has unit fallback cost. The lexicographic objective fixes the optimal or best-found aggregate fallback value before continuity is optimized. Among schedules having that same value, continuity may choose RF for an LC-preferring player or redistribute which player receives a fallback. The model does not separately optimize the maximum or distribution of individual fallback burden.

The eligible active set is

$$E_p(N) = \big(Q_p \cup h(Q_p) \cup \{\mathrm{C}\} \cup R_p(N)\big) \cap \mathcal{S}(N), \tag{1}$$

where

$$R_p(N) = \begin{cases} \{\mathrm{RC}\}, & N = 8 \text{ and } \mathrm{RF} \in Q_p, \\ \varnothing, & \text{otherwise.} \end{cases}$$

The special RF-to-RC rule applies only to the exactly eight-player defense, when both C and RF are inactive. It does not apply at nine players. Universal catcher eligibility is operational policy, while P and 1B are absent from $h(\cdot)$ and must be explicitly selected.

The fallback set is

$$F_p(N) = E_p(N) \setminus Q_p.$$

The model may assign p to F_p to preserve feasibility or fairness, but it minimizes such innings before evaluating any continuity term. Thus eligibility is broader than preference without erasing the semantic distinction between the two.

4 Constraint-Programming Formulation

For a fixed accepted roster size N, write $\mathcal{S}_N = \mathcal{S}(N)$, $K = K(N)$, $E_p = E_p(N)$, and $F_p = F_p(N)$. Table 2 collects the principal notation.

Table 2: Principal sets, variables, and aggregate quantities.

Symbol	Definition
$\mathcal{P}, \mathcal{I}, \mathcal{S}_N$	players, innings, and active positions
Q_p, E_p, F_p	explicit, eligible, and eligible non-explicit positions for p
x_{ips}, b_{ip}	field-assignment and bench-state indicators
t_p, d_p, Δ	innings played, scaled deviation, and playing-time spread
u_{ps}, v_p, e_p	position-use indicator, distinct positions, and excess above two
f	total eligible non-explicit assignment innings
o_1, o_2	numbers of exact one- and two-inning state runs
v, e, a	aggregate distinct positions, excess positions, and transitions
$H = 7K$	total fielding slots

4.1 Sets and decision variables

Let $\mathcal{P} = \{1, \dots, N\}$ be the available players and $\mathcal{I} = \{1, \dots, 7\}$ the innings. The primary binary variable is

$$x_{ips} = \begin{cases} 1, & \text{if player } p \text{ fields at position } s \text{ in inning } i, \\ 0, & \text{otherwise,} \end{cases}$$

for $i \in \mathcal{I}$, $p \in \mathcal{P}$, and $s \in E_p$. A second binary variable b_{ip} equals one exactly when player p is on the Bench in inning i. Omitting x variables for positions outside E_p makes an ineligible fielding assignment structurally impossible.

All variables denoted by x, b, w, a, $q^{(1)}$, $q^{(2)}$, and u are binary. The count variables satisfy $t_p \in \{0, \dots, 7\}$, $v_p \in \{0, \dots, K\}$, and $e_p \in \{0, \dots, \max(0, K-2)\}$.

Each active position is filled exactly once:

$$\sum_{p: s \in E_p} x_{ips} = 1 \qquad \forall i \in \mathcal{I},\ s \in \mathcal{S}_N. \tag{2}$$

Each player occupies exactly one field-or-Bench state:

$$b_{ip} + \sum_{s \in E_p} x_{ips} = 1 \qquad \forall i \in \mathcal{I},\ p \in \mathcal{P}. \tag{3}$$

Constraint (3) simultaneously prevents double assignment and explicitly represents the Bench. Because the Open profile has no gender rule, the model contains no gender-count or gender-placement constraint.

Before these variables are constructed, input validation rejects blank or duplicate names, empty or invalid preference sets, and out-of-range player counts. A distinct lineup preflight then considers the bipartite graph $G = (\mathcal{S}_N \cup \mathcal{P}, A)$, where $(s, p) \in A$ if and only if $s \in E_p$.

Proposition 1 (One-inning matching criterion)**.** *Under the Open profile, a legal seven-inning schedule satisfying only the hard assignment constraints exists if and only if G has a matching that covers $\mathcal{S}_N$. Equivalently, for every $T \subseteq \mathcal{S}_N$,*

$$\left| \bigcup_{s \in T} \{p \in \mathcal{P} : s \in E_p\} \right| \geq |T|.$$

Proof. Constraints (2)–(3) restricted to one inning are precisely a matching covering $\mathcal{S}_N$; Hall's theorem gives the stated equivalence. Availability and eligibility are invariant across innings, and the model has no cross-inning hard constraint. Repeating a covering matching therefore constructs a legal seven-inning schedule. Conversely, the assignment in any one inning of a legal schedule supplies such a matching. ∎

The implementation first reports any uncovered singleton position. Otherwise, it reports a deterministic minimum-cardinality Hall-deficient subset when no covering matching exists.

4.2 Playing-time quantities

The number of fielding innings received by player p is

$$t_p = \sum_{i \in \mathcal{I}} \sum_{s \in E_p} x_{ips}.$$

Let

$$t^{\max} = \max_{p \in \mathcal{P}} t_p, \qquad t^{\min} = \min_{p \in \mathcal{P}} t_p, \qquad \Delta = t^{\max} - t^{\min}.$$

There are $H = 7K$ fielding slots. To avoid fractional expressions, define the scaled deviation from an equal share as

$$d_p = |N t_p - H| .$$

Minimizing $\sum_p d_p$ is equivalent to minimizing total absolute deviation from H/N, up to the positive factor N.

4.3 State runs and transitions

For continuity purposes, each player has the state set

$$\mathcal{R}_p = E_p \cup \{\mathsf{B}\}.$$

Let w_{ipr} equal x_{ipr} for a field position $r \in E_p$ and equal b_{ip} for $r = \mathsf{B}$. A run is a maximal consecutive subsequence of identical states. Consequently,

$$\text{SS} \to \mathsf{B} \to \text{SS}$$

contains three runs and two transitions; it is not interpreted as one interrupted shortstop assignment.

For $i > 1$, a transition-start variable a_{ipr} represents the conjunction $w_{ipr} \wedge \neg w_{i-1,p,r}$ through

$$a_{ipr} \le w_{ipr}, \qquad a_{ipr} + w_{i-1,p,r} \le 1, \qquad a_{ipr} \ge w_{ipr} - w_{i-1,p,r}.$$

Exactly one new state starts whenever adjacent innings have different states. The first inning is not charged as a transition.

With boundary values $w_{0pr} = w_{8pr} = 0$, the binary one-inning-run indicator $q^{(1)}_{ipr}$, for $i \in \{1, \dots, 7\}$, recognizes the pattern $0, 1, 0$:

$$\begin{aligned} q^{(1)}_{ipr} &\le w_{ipr}, \\ q^{(1)}_{ipr} &\le 1 - w_{i-1,p,r}, \\ q^{(1)}_{ipr} &\le 1 - w_{i+1,p,r}, \\ q^{(1)}_{ipr} &\ge w_{ipr} - w_{i-1,p,r} - w_{i+1,p,r}. \end{aligned}$$

For $i \in \{1, \dots, 6\}$, the binary two-inning-run indicator $q^{(2)}_{ipr}$ recognizes the exact pattern $0, 1, 1, 0$ through

$$
\begin{aligned}
&q^{(2)}_{ipr} \le w_{ipr}, && q^{(2)}_{ipr} \le w_{i+1,p,r},\\
&q^{(2)}_{ipr} \le 1 - w_{i-1,p,r}, && q^{(2)}_{ipr} \le 1 - w_{i+2,p,r},\\
&q^{(2)}_{ipr} \ge w_{ipr} + w_{i+1,p,r} - w_{i-1,p,r} - w_{i+2,p,r} - 1.
\end{aligned}
$$

Binary domains make each linearization an if-and-only-if characterization. Both definitions apply to field positions and B. They measure rather than prohibit short runs; feasibility and higher-priority fairness remain paramount.

4.4 Positional variety

Let u_{ps} equal one if player p uses field position s at least once. The standard linkage is

$$
x_{ips} \le u_{ps} \quad \forall p \in \mathcal{P},\ s \in E_p,\ i \in \mathcal{I}, \qquad u_{ps} \le \sum_{i \in \mathcal{I}} x_{ips} \quad \forall p \in \mathcal{P},\ s \in E_p.
$$

Define the integer number of distinct field positions as

$$
v_p = \sum_{s \in E_p} u_{ps}.
$$

Mathematically, $e_p = \max\{v_p - 2, 0\}$. The implementation gives the nonnegative bounded integer e_p the constraints

$$
e_p \ge v_p - 2, \qquad e_p \ge 0.
$$

Because its objective coefficient is positive, the lower bound is tight in any minimizer of the continuity objective; equality is not imposed as a primitive constraint. An earlier incumbent retained after a time-limited phase need not carry a certificate of this tightness. Bench does not contribute to v_p or e_p. The two-position target is soft: the model may exceed it to protect legality, fairness, or the fallback count.

5 Lexicographic Objective

Define

$$
\begin{aligned}
f &= \sum_{i \in \mathcal{I}} \sum_{p \in \mathcal{P}} \sum_{s \in F_p} x_{ips},\\
o_1 &= \sum_{p \in \mathcal{P}} \sum_{r \in \mathcal{R}_p} \sum_{i=1}^{7} q^{(1)}_{ipr}, \quad & o_2 &= \sum_{p \in \mathcal{P}} \sum_{r \in \mathcal{R}_p} \sum_{i=1}^{6} q^{(2)}_{ipr},\\
e &= \sum_{p \in \mathcal{P}} e_p, & v &= \sum_{p \in \mathcal{P}} v_p,\\
a &= \sum_{p \in \mathcal{P}} \sum_{r \in \mathcal{R}_p} \sum_{i=2}^{7} a_{ipr}.
\end{aligned}
$$

Let $\boldsymbol{y}$ collect all decision variables, and let $\mathcal{X}$ be the feasible set defined by the preceding constraints. The ideal mathematical problem is

$$
\operatorname*{lexmin}_{\boldsymbol{y} \in \mathcal{X}} z(\boldsymbol{y}) = \left(\Delta, \sum_{p \in \mathcal{P}} d_p, f, o_1, e, o_2, v, a \right). \tag{4}
$$

Thus fairness has two ordered components, followed by the aggregate fallback count and five ordered continuity components.

The implementation scalarizes only within the fairness and continuity blocks. Since $0 \le t_p \le 7$, a valid bound is $D^{UB} = N \max\{H, |7N - H|\}$. The fairness objective is

$$J_F = (D^{UB} + 1)\Delta + \sum_p d_p. \tag{5}$$

The multiplier ensures that any one-unit reduction in spread dominates every possible change in total deviation.

After the fairness pair and fallback value have been fixed, the continuity objective is

$$J_C = c_1 o_1 + c_e e + c_2 o_2 + c_v v + a, \tag{6}$$

using the valid bounds

$$a^{UB} = 6N, \quad v^{UB} = NK, \quad o_2^{UB} = 6N, \quad e^{UB} = N \max\{0, K - 2\}$$

and the recurrence

$$\begin{aligned}
c_v &= a^{UB} + 1, \\
c_2 &= c_v v^{UB} + a^{UB} + 1, \\
c_e &= c_2 o_2^{UB} + c_v v^{UB} + a^{UB} + 1, \\
c_1 &= c_e e^{UB} + c_2 o_2^{UB} + c_v v^{UB} + a^{UB} + 1.
\end{aligned}$$

Proposition 2 (Lexicographic equivalence)**.** *Minimizing J_C is equivalent to lexicographically minimizing (o_1, e, o_2, v, a) over any feasible slice on which the fairness pair and fallback value are fixed.*

Proof. Each coefficient exceeds the largest possible contribution of all terms to its right. Consequently, a unit improvement in the first differing component strictly decreases J_C, irrespective of every lower-priority component; the converse follows by applying the same domination argument to the first differing component of two feasible vectors. ■

It follows that continuity cannot increase the fixed fallback value, although it may redistribute the same number of fallback assignments among players. A B run is evaluated by the same logic as a field-position run. Exact two-inning runs are minimized only after one-inning runs and excess positions; the model therefore never purchases a reduction in o_2 by worsening o_1 or e.

6 Lexicographic Solution Method

The model is implemented with the OR-Tools CP-SAT solver, which supports Boolean and integer constraint models and distinguishes statuses including *OPTIMAL*, *FEASIBLE*, and *INFEASIBLE* [1]. The interactive default is a nominal five-second multi-phase solve target measured only after model construction. It is not a hard end-to-end wall-clock bound: model-building time, Python and Streamlit overhead, CP-SAT time-limit behavior, and the implementation's minimum per-phase and subphase allocations of 0.05–0.1 seconds can make observed elapsed time exceed five seconds. The implementation uses the following nominally bounded phases:

1. **Balanced-count feasibility test.** Write $H = qN + r$, where $0 \le r < N$. Without eligibility restrictions, the balanced count multiset has $N - r$ values of q and r values of $q + 1$, with $\Delta = \mathbf{1}_{\{r>0\}}$ and $\sum_p d_p = 2r(N - r)$. Add $\lfloor H/N \rfloor \le t_p \le \lceil H/N \rceil$ for every player and test feasibility. A feasible assignment certifies that both arithmetic lower bounds remain attainable under all hard constraints.

2. **General fairness search.** If that slice cannot be established, minimize J_F within the designated fairness budget.
3. **Preference phase.** Fix the achieved spread and deviation, minimize f, and fix the exact fallback incumbent before any continuity phase. If optimality is not proved, the incumbent is still frozen rather than silently relaxed.
4. **One-inning phase.** For at most 0.25 seconds, test the lower-bound slice in which the only single-inning field run is forced by $t_p = 1$ and the only single-inning B run is forced by $t_p = 6$. Within this hard slice, minimize e as look-ahead to the next tier. If feasibility is not established within that time slice, minimize $o_1(E^{UB} + 1) + e$ for up to 2.5 seconds, where E^{UB} bounds total excess positions, and preserve the one-inning incumbent. If that search produces no candidate, retain the already legal fairness/fallback incumbent and continue with the complete continuity objective. The coefficient keeps every unit of o_1 strictly more important than all possible changes in e.
5. **Two-position phase.** If the preceding o_1 tier is proved, test the $e = 0$ slice inside the existing phase budget and otherwise minimize e with the unspent remainder. A proved value is fixed. If o_1 remains unproved, do not independently cap e: retain the complete ceiling $J_C \leq J_C^{inc}$, where J_C^{inc} is the accepted incumbent's integer objective value, so a later improvement in o_1 may legitimately spend lower-priority quality.
6. **Two-inning phase.** When o_1 and e are both proved, minimize $o_2c_2 + vc_v + a$; otherwise keep the complete J_C objective active. Run two concurrent four-worker searches sharing the existing eight-worker computational envelope. One is an ordinary proof-capable CP-SAT run; the second is a hint-seeded large-neighborhood search with a different seed. Retain the lexicographically better legal candidate. Fix o_2 only when its prefix is proved; otherwise preserve the full weighted incumbent ceiling.
7. **Tail-continuity phase.** Use the remaining nominal budget to minimize $vc_v + a$ only when every earlier continuity tier is proved; if not, continue minimizing J_C. Because a hint is not a quality guarantee, constrain the complete weighted objective to be no larger than the incumbent and accept a timed candidate only when its full (o_1, e, o_2, v, a) vector is lexicographically no worse than that incumbent.

An *UNKNOWN* status from a time-limited slice is not interpreted as infeasibility. Every candidate is required to have a solved status and is then compared with the accepted incumbent by tuple lexicographic order, not componentwise order. A proved higher-priority value is fixed before the next phase begins; without such a proof, the complete incumbent ceiling remains active.

The application synthesizes its public status from all phase results. It reports *OPTIMAL* if and only if every prefix proof flag is true and the final tail solve returns CP-SAT *OPTIMAL*. Otherwise any returned legal incumbent is labeled *FEASIBLE*, even when one or more prefix tiers are proved. The latter label denotes absence of a complete lexicographic optimality certificate, not uncertainty about legality.

7 Computational Evidence

Here For The Beer (HFTB) is a separate fifteen-player co-ed benchmark instance; it is not Team Red evidence. On the recorded macOS reference environment at commit `fa92000`, twenty measured HFTB calls to `optimize_game` were distributed across four fresh Python processes after one excluded warm-up per process. The reported duration includes model construction and all solver phases, but excludes process startup, roster loading, and the excluded warm-up. All twenty preserved the proved fairness/fallback prefix $(4, 210, 0)$. Nineteen met the timed continuity target $(5, 0, 9, 20, 21)$ lexicographically: twelve returned $(5, 0, 9, 19, 20)$ and seven returned $(5, 0, 9, 20, 20)$. One legal incumbent returned $(5, 0, 10, 20, 21)$ and is retained explicitly in the evidence rather than hidden by an aggregate. Call duration had p95 5.062 seconds and maximum 5.063 seconds; the fairness/fallback prefix had p95 0.138 seconds and maximum 0.151 seconds. Every result remained *FEASIBLE*. The configured experimental gate was an acceptance proportion of at least 0.90; the observed empirical proportion was $19/20 = 0.95$. No population guarantee or confidence bound is claimed. The p95 statistic is the nearest-rank order statistic for $n = 20$, not an estimate of a service-level guarantee.

Table 3: Recorded HFTB benchmark results at revision `fa92000`.

Quantity	Recorded value
Measured repetitions	20
Certified fairness/fallback prefix	$(4, 210, 0)$ in 20/20 runs
Continuity vector $(5, 0, 9, 19, 20)$	12 runs
Continuity vector $(5, 0, 9, 20, 20)$	7 runs
Continuity vector $(5, 0, 10, 20, 21)$	1 run
Target-satisfying proportion	$19/20 = 0.95$
Call duration, p95 / maximum	5.062 / 5.063 seconds
Prefix duration, p95 / maximum	0.138 / 0.151 seconds

8 Team Red Case Study

This case study uses the fourteen-player Team Red snapshot reproduced in Table 4, with every player initially available under locked Open rules.

Table 4: Team Red roster snapshot and explicit positional preferences.

Player	Explicit preferences	Player	Explicit preferences
David R	2B, 3B, RC	Justin	1B, RC
Andrew	1B, 2B	Kevin	2B, LC
David	RF	Dung	P
Heather	LF, LC	Ryan	1B, 3B
Tristan	C, 3B	Brad	C, LF
AP	SS, LF	Chris	C, 2B, RF
Marty	SS, LC	Shaz	C, RF

Table 5: Representative Team Red defensive schedule.

Inning	P	C	1B	2B	3B	SS	LF	LC	RC	RF
1	Dung	Brad	Andrew	Chris	Ryan	Marty	Heather	Kevin	David R	David
2	Dung	Brad	Justin	Chris	Ryan	Marty	Heather	Kevin	David R	David
3	Dung	Shaz	Justin	Chris	Tristan	AP	Heather	Kevin	David R	David
4	Dung	Shaz	Justin	Andrew	Tristan	AP	Heather	Kevin	David R	David
5	Dung	Shaz	Ryan	Andrew	Tristan	AP	Brad	Marty	David R	David
6	Dung	Shaz	Ryan	Andrew	Tristan	AP	Brad	Marty	Justin	Chris
7	Dung	Shaz	Ryan	Andrew	Tristan	AP	Brad	Marty	Justin	Chris

Table 5 records one machine-checked incumbent produced by the nominal five-second procedure. It is included as a primal feasibility witness, not as a claim of uniqueness.

Proposition 3 (Optimal Team Red fairness profile)**.** *For the snapshot in Table 4, every feasible schedule has $\Delta \geq 3$. Conditional on $\Delta = 3$, the minimum scaled absolute deviation is 56, attained by the inning-count multiset $\{7^1, 5^{11}, 4^2\}$.*

Proof. Here $N = 14$, $K = 10$, and $H = 70$. Pitcher is active in every inning, and Dung is the unique player whose eligible set contains P. Let p_D denote Dung. Hence $x_{ip_D\mathrm{P}} = 1$ for every $i \in \mathcal{I}$ and $t_{p_D} = 7$. If $\Delta \leq 2$, every other player must receive at least five innings, implying $\sum_p t_p \geq 7 + 13(5) = 72 > H$, a contradiction. Thus $\Delta \geq 3$.

The schedule in Table 5 has count multiset $\{7^1, 5^{11}, 4^2\}$ and therefore establishes $\Delta \leq 3$. For the second tier, let a_j count players with j innings in a spread-three schedule. Because $4 \leq t_p \leq 7$, $a_4 + a_5 + a_6 + a_7 = 14$ and $4a_4 + 5a_5 + 6a_6 + 7a_7 = 70$, whence $a_4 = a_6 + 2a_7$. Moreover $a_7 \geq 1$, and

$$\sum_p d_p = 14a_4 + 14a_6 + 28a_7 = 28a_6 + 56a_7 \geq 56.$$

The displayed schedule attains 56, completing both bounds. ■

For the complete vector in (4), the displayed incumbent has

$$z = (3, 56, 0, 3, 0, 15, 20, 20).$$

The zero fallback count is optimal conditional on the certified fairness pair because $f \geq 0$. The application status was *FEASIBLE*, so no optimality claim is made for $(o_1, e, o_2, v, a) = (3, 0, 15, 20, 20)$. Multiple name-by-name schedules may have the same vector.

The recorded run used code revision `ff382de`, roster SHA-256 `4b3f2715e9dbb40dc8dfac945ea0 60953f6c31b17fd6d8be2d94af0ff33b9ad3`, Python 3.9.6, OR-Tools 9.15.6755, and macOS 26.6.2 on arm64. The measured `optimize_game` call took 5.073 seconds under the nominal five-second budget. This single recorded execution is a reproducible artifact datum, not a sampling statement about latency or incumbent quality.

The instance also exposes availability risk clearly. If Dung is marked unavailable, no remaining player has opted into P. Simultaneous absences by Justin, Andrew, and Ryan leave eleven players but remove all 1B coverage, while simultaneous absences by AP and Marty leave twelve players but remove all SS coverage. Adequate headcount therefore does not guarantee positional feasibility. The correct response is an infeasibility message identifying the uncovered position, not an invented assignment and not a reduced nine-player lineup. The absence of any catcher opt-in, by itself, cannot leave C uncovered because $\mathrm{C} \in E_p$ for every player; another Hall conflict could still make the lineup infeasible.

9 Verification and Falsification Strategy

Validation is organized around properties rather than visual plausibility. At prepublication revision `ff382de`, a local run recorded 144 passing pytest cases. The Playwright suite defines eight scenario functions, each configured for a mobile-touch project and a desktop-mouse project, yielding 16 project-test executions. These are implementation-conformance checks; they are not substitutes for the analytic bounds or solver certificates stated above. The principal checks include:

1. **Legality invariants:** seven innings, exact active-position coverage, no duplicated player within an inning, valid eligibility, and reported summary counts recomputed from returned assignments, with separate assertions for eligibility.

2. **Open-profile boundaries:** legal eight-, nine-, and ten- fielder position sets; no gender error in Open; no nine-player RF-to-RC special promotion; universal C fallback; and explicit P/1B requirements.
3. **Team Red contract:** exact names and preferences, internal `Unspecified` gender, locked Open rules, all players initially available, fair inning distribution, and no player assigned to more than two field positions.
4. **Preference-priority witness:** the fixture implemented by `test_explicit_preferences_beat_a_strictly_smoother_fallback_schedule` in `tests/test_optimizer.py` has a solver-proven seven-fallback optimum with continuity vector $(5, 1, 16, 20, 21)$. Promoting Third Specialist's already-eligible 2B from a fallback to an explicit preference leaves the eligibility sets unchanged and yields the smoother vector $(5, 0, 14, 20, 20)$. Rescored under the original preferences, that exhibited schedule uses nine fallbacks. The five vector components are, respectively, one-inning runs, excess positions, two-inning runs, total distinct positions, and adjacent transitions; the first improvement is excess positions, from one to zero. The original optimizer nevertheless selects the seven-fallback result. The test detects violations on this fixture; the general non-worsening property follows from imposing the exact constraint $f = f^{inc}$ before continuity search.
5. **Mutation test:** weakening the exact fallback-incumbent freeze to a one-sided inequality is detected. The regression inspects the model domain and requires the exact singleton interval $[f, f]$.
6. **Interaction tests:** real individual position taps, complete name entry, gender change where applicable, save/reopen persistence, optimization, the complete semantic seven-inning matrix, CSV download, invalid optimization recovery, removal, reset, and independent browser sessions.

The strict preference witness is important because a test that merely observes zero fallback use is vacuous: continuity cannot demonstrate an illegal trade when no trade is available. The constructed counterexample supplies a smoother but preference-worse alternative and verifies that the implemented priority rejects it. This is closer to falsification than to screenshot confirmation.

10 Human and Operational Interpretation

The objective order encodes several practical slowpitch considerations. Continuous field-position blocks reduce repeated instruction, warm-up changes, and uncertainty between innings only as an intended operational proxy; no user study estimates those causal effects. Modeling B as a state prevents the objective from ignoring repeated movement between the field and bench. Consecutive bench innings are likewise intended to simplify communication from a phone in a dugout, but that usability claim has not been experimentally evaluated.

The hierarchy supplies controlled resilience without pretending all positions are interchangeable. Its SS-to-3B/2B, 3B-to-2B, and LC-to-LF/RC/RF pathways are declared team-policy eligibility rules, not universal claims that one position is easier than another. The separate fallback phase prevents continuity from increasing the fixed aggregate fallback value, while still permitting redistribution among schedules tied at that value. Pitching and first base remain explicit-only choices.

Operational authority is scoped to an input snapshot, not to every open web session. Team Red's defaults mark all fourteen players available, but that is not attendance confirmation. Before solving, one designated lineup caller must mark every scratch. Any later availability or preference change requires a new solve, and the newly displayed or downloaded matrix supersedes the previous one without hand editing. Because browser sessions are independent and not synchronized, the team should distribute the result from that single source, preferably by downloading the final matrix before the first pitch.

No captain-review stage follows optimization. Such a stage would blur the contract by presenting a computed schedule and then asking a human to repair it without rechecking legality, fairness, or continuity. In the implemented workflow, changed judgments are expressed as changed availability or preferences and submitted to a new solve. The optimizer, rather than an unvalidated hand edit, remains the source of the displayed and downloaded lineup.

11 Limitations and Threats to Validity

The model is intentionally narrow. Bench means not fielding; it does not alter batting order. The model does not optimize offensive substitutions, opponent tendencies, score-dependent tactics, player fatigue, or inning-specific availability. It balances playing time within one game and does not carry season-to-date equity between sessions. A shortened game simply leaves later rows unused and may make realized playing time less equal than the planned seven-inning distribution; extra innings are outside the fixed horizon. The hierarchy is a declared operational policy, not an empirical estimate of every individual's defensive ability.

Universal catcher eligibility is likewise a policy assumption. The software correctly applies it to every roster and does not offer a catcher opt-out. If a future league or team rejects that assumption, the input schema and hard eligibility rules would need an explicit change; silently weakening the rule inside the solver would be inappropriate.

The objective treats all explicit preferences as equal and all eligible non-explicit assignments as equal. It does not know that one player strongly favors LC over LF unless the roster is edited to reflect only the intended choices. A future ranked-preference model could add tiers, but it would require additional input and a revised priority contract.

Finally, each CP-SAT phase is assigned a nominal time limit. A *FEASIBLE* result is guaranteed to satisfy the encoded hard constraints but may lack a proof that every reported objective tier is globally best. Conservative status reporting makes this uncertainty visible, and the web interface reports observed elapsed time. This paper does not infer latency percentiles or a hard wall-clock guarantee from the mere existence of a benchmark utility. Any expansion beyond the eight-to-fifteen-player scope or seven-inning horizon requires renewed, reproducible timing measurement.

12 Conclusion

Open-division defensive scheduling is best viewed as a lexicographic state- assignment problem. The model fills every required position, equalizes playing time as far as eligibility permits, preserves explicit preferences before using controlled eligibility expansion, and then optimizes ordered continuity measures over field and B states. Fixing the aggregate fallback value before continuity search preserves the distinction between explicitly declared positions and merely eligible alternatives, although individual fallback assignments may still differ among aggregate ties.

For Team Red, Proposition 3 proves a minimum playing-time spread of three and a minimum scaled deviation of 56. The recorded incumbent uses no fallback assignments, assigns no player to more than two field positions, and has continuity vector $(3, 0, 15, 20, 20)$; the public *FEASIBLE* status expressly withholds a claim of complete continuity optimality. The staged procedure is therefore suitable as a legal-incumbent generator on an interactive time scale, subject to the empirical and modeling limitations stated above. Natural extensions include inning-specific availability, season-level equity, ranked preferences, and empirical calibration of player–position suitability.

A Algorithmic Summary

The implemented procedure may be summarized as follows:

1. normalize and validate the selected available players;
2. select the Open active-position set from N;
3. derive E_p and F_p while preserving Q_p;
4. preflight a distinct-player matching that covers all active positions;
5. construct field, B, fairness, run, position-use, and transition variables;
6. attempt the balanced-count feasibility certificate, otherwise optimize fairness;
7. constrain the achieved fairness pair exactly;
8. minimize and freeze fallback innings;
9. minimize and preserve one-inning state runs;
10. minimize excess positions and prevent subsequent worsening;
11. minimize the lexicographically weighted continuity expression;
12. translate the legal incumbent into an inning matrix and player summary; and
13. label the result *OPTIMAL* only when all required proofs exist, otherwise label the legal result *FEASIBLE*.

B Application and Reproducibility Artifacts

The public neutral application is available at `https://softball-fielding-optimizer.streamlit.app/`. It presents an explicit first-load choice among Here For The Beer, Team Red, and a blank roster. Team Red loads the roster in `team_red_roster.csv` under locked Open rules. The existing Here For The Beer deployment remains available at `https://here-for-the-beer.streamlit.app/` and retains its Co-ed defaults.

The implementation, normative specification, test suite, and benchmark utilities are maintained in the public repository at `https://github.com/davidluizrusso/softball-fielding`. Streamlit Community Cloud deploys an application from a GitHub repository and entrypoint and provides a shareable `streamlit.app` URL [3]. Python dependencies are declared in `requirements.txt`, consistent with the platform's dependency mechanism [4].

During a solve, the web interface reports the active phase and displays a small animated stick figure. This cue is intended to distinguish active computation from an unregistered input event; no usability experiment is claimed.

The command-line verification entry points are:

- `python -m pytest` for model and Streamlit state tests;
- `npm run test:browser` for mobile-touch and desktop-mouse Playwright workflows;
- `python scripts/benchmark_fairness.py -budgets 1 2 3` for repeated fairness-phase stress trials at selected roster sizes; and
- `python scripts/benchmark_hftb.py -processes 4 -runs-per-process 5 -budget 5 -output benchmarks/hftb-5s-reference.json` for the opt-in 20-run HFTB latency, phase-timing, and lexicographic-quality contract across fresh processes.

The PDF was built with pdfTeX 3.141592653-2.6-1.40.29 (TeX Live 2026) by running `pdflatex` twice with `-interaction=nonstopmode` and `-halt-on-error`. Exact Python and OR-Tools versions are recorded for the Team Red witness because the version ranges in `requirements.txt` are deployment constraints rather than a fully pinned research environment.

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